

UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

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COURSE CODE

SECV 2113 – Human Computer Interaction

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SECTION

Section 1

(a) Purpose of TNGo

Identified Purpose: **TNGo** is designed to simplify digital financial transactions, helping users manage their finances easily and securely. By providing rewards, cashbacks, and financial insights, it aims to promote better financial habits and empower users to take control of their spending. **TNGo** is a mobile app designed to facilitate cashless transactions and digital payments, promoting financial inclusion and economic growth in underserved communities. Its main purpose is to provide a secure, efficient, and user-friendly payment solution that encourages the use of technology in daily financial activities, reducing the reliance on cash and traditional banking systems. This app helps bridge the gap between the unbanked population and financial services, fostering innovation and building resilient infrastructure.

Sustainable Development Goal: **TNGo** aligns with SDG 9 (Industry, Innovation, and Infrastructure) by promoting financial inclusion and supporting digital infrastructure, helping more people access convenient cashless solutions. **TNGo** is a mobile app that promotes financial inclusion and economic growth in underserved communities by facilitating cashless transactions and digital payments. It aligns with Sustainable Development Goal 9: Industry, Innovation, and Infrastructure by building resilient financial infrastructure, fostering innovation, enhancing resource-use efficiency, and supporting sustainable development, ultimately enabling even the most marginalized communities to participate in the digital economy. By enabling cashless transactions, **TNGo** helps promote inclusive industrialization. It allows small and medium-sized enterprises (SMEs) to participate in the digital economy, fostering economic growth and creating job opportunities.

(b) Analysis of Two Features Using the FBM

Feature 1: In-App Rewards and Cashback

Description:

The In-App Rewards and Cashback feature incentivizes users to utilize the **TNGo** app for their transactions by providing monetary benefits for specific purchases or payment activities. This encourages increased engagement with the app as users can earn rewards or cashback on their everyday transactions. Encouraging more transactions leads users to consider **TNGo** as their go-to payment solution for both small and large purchases. This change in consumer behavior can make the app central to users' financial transactions, thus increasing user retention rates.

Target Behavior:

The primary target behavior is to drive users to make more transactions through the TNGo app, thereby enhancing user loyalty and increasing transaction volume. The immediate gratification of earning rewards or cashback serves as a psychological trigger. As users experience the benefits of these rewards, they are likely to continue using the app, reinforcing a cycle of behavior that results in even more transactions. As users see their rewards grow, they may feel compelled to continue using the app to maintain or surpass their friends' rewards, creating a friendly competitive environment. This social aspect can be a powerful motivator for users who enjoy sharing achievements.

Motivation:

The promise of rewards or cashback offers immediate value, appealing to users' pleasure and hope for saving money on regular transactions: The allure of earning rewards or cashback provides users with tangible benefits. This taps into the pleasure principle, where users feel rewarded for their spending, leading to a positive reinforcement cycle. For example, they buy RM5 cash voucher with MYR4.10+5Pts, they can save their money and continue earning point by claiming daily rewards. Cashback offers can be tailored to specific spending categories (like groceries, dining, or entertainment), aligning with users' typical purchasing behavior. When users receive cashback on transactions they would make anyway, it reinforces their decision to continue using TNGo as their payment platform of choice.

Social Influence: If rewards are shareable or can be tracked on social platforms, users may feel motivated to earn and showcase their rewards to peers. This fosters a sense of competition and social acceptance, where users want to compare their rewards against those of friends or family. They can also share the reward to their friend like they can send as a gift. When friends or family share their positive experiences of earning cashback through TNGo, it reinforces the app's value proposition. Users may feel more inclined to use TNGo for their transactions if they see their peers benefitting from it. If users can track and share their rewards with others, this social aspect enhances motivation. Users may want to show off their cashback earnings, which can create a sense of community and encourage others to join in.

Savings Appeal: Users are naturally inclined to save money; thus, the prospect of receiving cashback on regular transactions aligns with their financial motivations. Cashback offers an immediate return on transactions. Users

appreciate the feeling of getting something back with each purchase, reinforcing positive associations with the act of spending. This instant gratification can drive them to choose TNGo over competing payment methods that do not offer similar rewards. The ability to earn cashback creates a sense of accomplishment and satisfaction. Users feel they are "winning" by receiving money back on their purchases, which enhances their overall experience and encourages continued use of the app.

Ability:

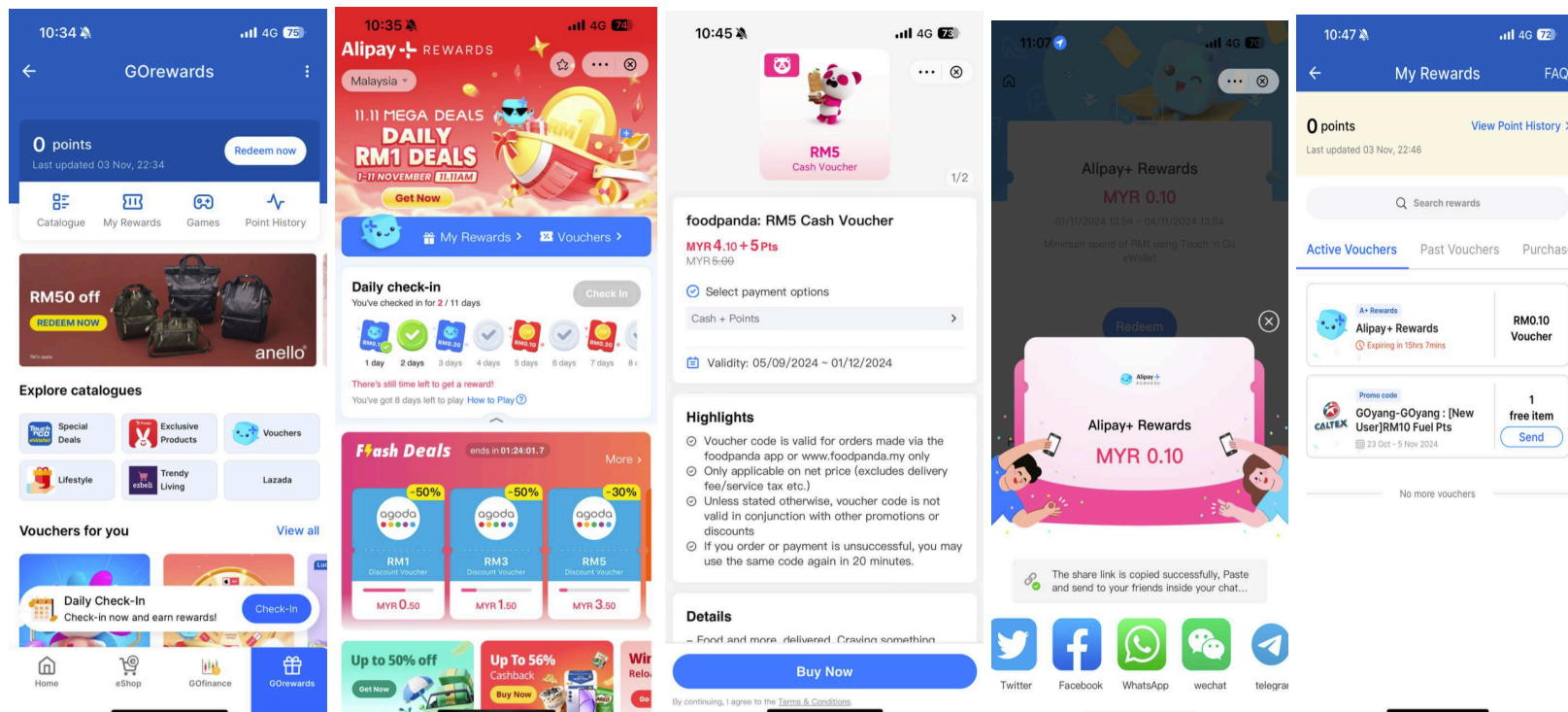
The TNGo app is designed with simplicity in mind, minimizing the number of steps required to earn rewards. Users can easily view available rewards and their progress without navigating through complex menus. User enter the interface, it click check in to claim daily check in, you can choose reward what you want based on filter and category, such as food and beverage, taobao, etc. The app clearly communicates how users can earn rewards, making it straightforward for them to understand the system. You can check what you have in the My Rewards, it is Streamlined Processes. The app is designed to have streamlined processes for earning rewards, such as straightforward transaction flows. Users can earn rewards with minimal input, reducing the frustration often associated with more complex systems. Key information about available rewards is prominently displayed on the main interface, ensuring that users can quickly identify opportunities to earn benefits. This visual hierarchy helps direct user attention where it is needed most, making it easier to engage with the rewards system. TNGo clearly outlines how users can earn rewards. For instance, it specifies the percentage of cashback available for different categories or explains any conditions (such as minimum transaction amounts) in simple terms. This transparency eliminates confusion and builds user trust in the rewards program. Users receive real-time notifications regarding their rewards status, such as when they earn cashback or when new opportunities become available. These notifications keep users informed and engaged, prompting them to take action and make transactions through the app.

Trigger:

The app sends timely notifications to users, informing them when they are eligible for cashback or rewards on upcoming transactions. For instance, if a user is near a participating retailer that offers a limited-time cashback promotion, the app can send a push notification alerting them to the promotion. This timely reminder acts as a

powerful trigger that can lead to immediate purchases, leveraging the principle of urgency. Limited-time offers create a sense of urgency, tapping into the psychological principle of scarcity. When users know that an opportunity to earn cashback is available for only a short period, they are more likely to prioritize transactions through the app to avoid missing out. Users can also receive notifications when they reach certain milestones, such as accumulating a specific amount of rewards, encouraging them to make additional transactions to reach the next reward tier. Users can receive alerts about how close they are to reaching the next reward tier or bonus. For example, a notification might say, “You’ve checked in for 2/11 days” This creates a clear and enticing goal for users to aim for, motivating them to make additional transactions to reach that target. Ultimately, these triggers play a crucial role in reinforcing the rewards system and promoting ongoing user loyalty to the TNGo platform, creating a dynamic cycle of engagement that benefits both users and the app itself.

Supporting Image: with labels highlighting the trigger (notification), motivation (rewards displayed), and ability



Feature 2: Expense Breakdown and Money Flow Overview

Description: The Expense Breakdown and Money Flow Overview feature provides users with a comprehensive view of their financial activities, displaying detailed insights into where their money is being spent. This feature aggregates transaction data, categorizes expenses, and presents it visually, enabling users to understand their spending patterns better and make informed financial decisions. The goal is to empower users to manage their budgets more effectively by highlighting areas of high expenditure and identifying opportunities for savings.

Target Behavior: The primary target behavior for this feature is for users to check their expense breakdown regularly. By engaging with this feature consistently, users can improve spending habits which regularly reviewing their expenses encourages users to reflect on their spending patterns. This awareness can lead to conscious decision-making regarding future purchases, helping them to align their spending with their financial goals. They can achieve financial awareness. Users who actively monitor their expenses are more likely to develop a deeper understanding of their financial situation. This awareness can foster a sense of accountability, prompting them to adjust their habits accordingly to avoid overspending. They can engage with budget management. By analyzing their expenses, users can identify categories where they may be overspending, enabling them to set realistic budgets for those categories. This proactive approach to budget management can lead to more disciplined financial behavior over time. By visualizing both income and expenses, users can identify trends in their spending behavior, enabling them to make informed decisions about where to cut back or reallocate funds.

Motivation:

Users are motivated by hope to improve financial stability, and possibly fear of overspending. Knowing where their money goes provides control, appealing to users' long-term goals. The fear of falling into debt or overspending can be a powerful motivator. When users have visibility into their spending habits, they can take steps to curb unnecessary expenses and prevent financial stress, appealing to their desire for control over their finances. Users are often motivated by the hope of achieving greater financial stability. By

understanding their expenses, they can make informed choices that contribute to their long-term financial well-being. The potential to save money and build a more secure financial future drives users to engage with the app. Users are motivated by their long-term financial aspirations, such as saving for a home, retirement, or a major purchase. Knowing where their money goes can help them align their spending with these goals, making the budgeting process feel more purposeful and achievable. Users are motivated by the need for clarity in their financial situation. Understanding both income and expenses gives them a complete picture, helping them feel more in control of their finances and reducing anxiety about money management.

Ability:

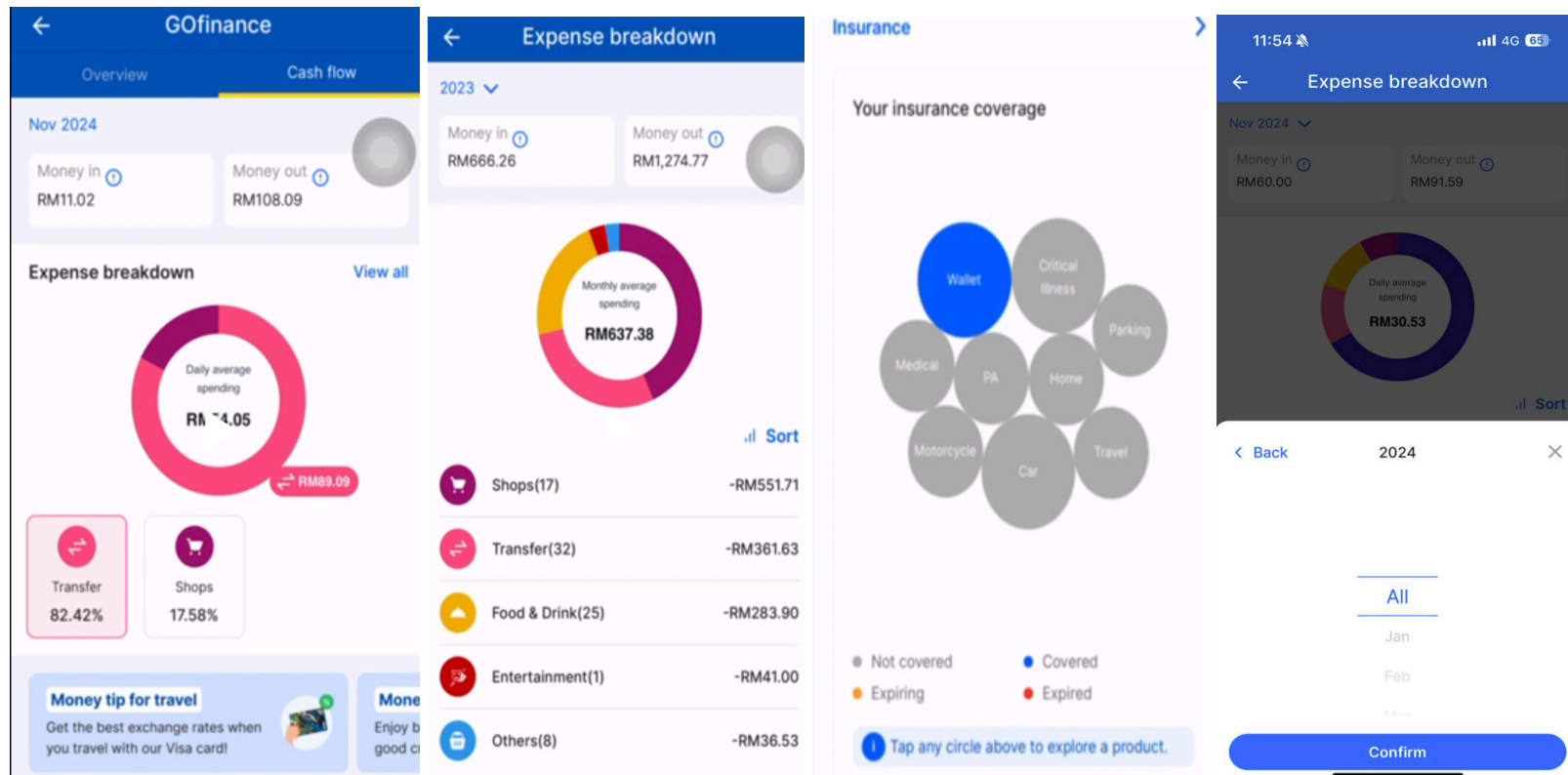
The app presents expense information in an easily digestible format. By using intuitive layouts and straightforward categories (e.g., shops, entertainment, transfer), users can quickly understand their spending without feeling overwhelmed by complex data. Users don't need to perform manual calculations, which reduces brain effort and makes insights immediately accessible. The inclusion of visual aids like charts and graphs simplifies the representation of financial data. For example, a pie chart illustrating the percentage of total spending by category allows users to visualize their money flow at a glance, enhancing their comprehension of financial patterns. The app performs all necessary calculations automatically, freeing users from the burden of manual tracking and analysis. This ease of access makes financial insights immediately available, enabling users to make informed decisions with minimal effort.

Trigger: Monthly Spending Summary on the Go Finance

TNGo can implement an in-app widget that appears on the daily average spending within the expense overview section. This widget displays a of the previous month's spending or any month or overall year spending via filter , highlighting insights into high-expense categories and overall spending trends. This signal acts as a natural monthly summary, prompting users to engage with their expense data at the beginning or previous of each month when they are likely reviewing finances. This unobtrusive trigger provides a consistent check-in without relying on external reminders. Users are encouraged to explore their spending data at a time when they are already focused on their finances, increasing the likelihood of engagement. It also show cash flow of money in and money out. By presenting cash flow information clearly,

the app encourages users to reflect on their financial situation and make necessary adjustments for the upcoming month. This proactive approach fosters a habit of financial awareness and responsibility.

Supporting Image: expense overview feature, labeling the trigger (notification), motivation (graphical expense summary), and ability (simple, categorized display).



c) https://youtu.be/Tshh8y1Lt_Q

D) Failed Feature Analysis and Improvement Suggestions

Identified Failed Feature: Ineffective necessary Face Recognition for Login

Description: TNGo has implemented a face recognition feature for login purposes, aiming to enhance user convenience by allowing quick access to the app. However, if this feature is unreliable—evident through slow response times or difficulty recognizing faces under different lighting conditions or angles—it can lead to user frustration. Instead of providing a seamless experience, the feature may inadvertently create barriers to entry, dissuading users from utilizing the app. Users would be unable to access their stored funds within the e-wallet, which could prevent them from completing essential transactions, such as paying bills, making purchases, or transferring money to others. This can lead to frustration and financial inconvenience.

Why It Fails (Based on FBM):

Ability: The FBM states that ability refers to how easy or difficult a behavior is to perform. When face recognition fails frequently or requires multiple attempts, it undermines the simplicity and convenience intended for the login process. Users expect a quick and hassle-free experience; if they must struggle to log in, the perceived effort increases significantly. Users encountering repeated failures may begin to feel that accessing the app requires more effort than anticipated. This can lead to avoidance of the face recognition feature altogether, opting instead for traditional login methods, which may feel less modern or convenient. Users who are less tech-savvy may struggle with using face recognition features. If the user interface is not intuitive or requires prior knowledge, it can impede their ability to log in successfully. External factors such as poor lighting or background distractions can significantly affect face recognition performance. Users may find it challenging to log in from various environments if the system does not adapt to these conditions. Not all devices have advanced camera capabilities for accurate face recognition. Users with older devices may experience more login failures, limiting their ability to utilize this feature effectively. The absence of accessibility features for users with disabilities (such as visual impairments) can restrict their ability to use the app. Providing alternative login methods or voice commands could enhance accessibility. If the app is slow to process face recognition, users may become impatient. A lengthy response time can diminish their ability to log in quickly, leading to frustration and disengagement.

Motivation: The face recognition feature is designed to provide a seamless and efficient login experience. However, repeated failures diminish users' motivation to rely on this feature. If the system frequently fails to recognize them, users may feel disheartened and less inclined to attempt using face recognition again, leading to a loss of trust in the feature. When motivation decreases due to frustration with a feature, users may disengage from the app altogether, impacting overall user retention and satisfaction. Recommendations from friends or family can increase motivation. If users see others successfully using the face recognition feature, they may be more inclined to try it themselves.

Trigger: The face recognition prompt acts as a trigger each time a user attempts to log in, ideally facilitating quick access to the app. However, if the feature does not work smoothly and users find themselves needing to log in multiple times, this trigger can become a source of frustration instead of an invitation to engage. Sudden logouts or failures to recognize the user can interrupt the flow of their intended interaction, leading to a negative experience while in urgent to pay the money and potential abandonment of the app. Including visual indicators, such as a friendly animation or encouraging message on the login screen, can make the face recognition prompt feel more inviting. Aesthetic elements can positively influence the trigger's effectiveness. Implementing immediate feedback after a login attempt (success or failure) can serve as a trigger for users to take further action. For instance, if a login fails, providing clear next steps or alternative options can prompt users to continue trying to access their accounts.

Improvement Suggestions:

We can enhance Face Recognition Accuracy. Invest in improving the face recognition software to perform reliably across various lighting conditions and angles. Advanced AI algorithms can be employed to enhance recognition accuracy and speed, making the feature more dependable. Implement machine learning techniques that allow the system to learn from user interactions, improving its ability to recognize users in different environments over time. We can provide alternative options. If face recognition fails after a couple of attempts, automatically prompt the user with a secondary option, such as a PIN or fingerprint login, to reduce friction. We inform user that they can switch to an alternative login method if face recognition fails, reinforcing that their access is still secure and convenient. Only activate the face recognition prompt when the user's device is at an optimal distance for recognition. Providing on-screen guidance or visual indicators for proper positioning can significantly increase the chances of successful recognition on the first attempt. Incorporate immediate feedback mechanisms that let users know if they are too close, too far, or at an incorrect angle, thereby improving user experience during the login

process. Strengthen security features to prevent unauthorized access while ensuring that legitimate users can log in smoothly. Multi-factor authentication (MFA) can provide an additional layer of security without sacrificing usability. Ensure the app is regularly updated to address bugs and improve performance. Regular maintenance can help prevent issues that lead to login failures. Develop a responsive customer support system that can assist users quickly in the event of login failures. Consider live chat options or an FAQ section specifically addressing login issues. Encourage users to provide feedback on their login experiences. Regularly analyze this feedback to identify patterns and areas for improvement, allowing for continuous enhancement of the login process.

Supporting Image: showing face but no face detected, showing face but appear bugs and log out of tngo

